A Reduction of the Squarefree Coprime Adjacent Divisor Problem and a Conditional Golden-Ratio Lower Bound

Michael M. Ross
Independent Researcher
michaelmross@cantab.net

Abstract

For squarefree n with $\omega(n) = k$, let $\tau_\perp(n)$ be the number of adjacent pairs in the divisor ordering of n that are coprime, and let $g_{\mathrm{sf}}(k) = \max_{\omega(n)=k} \tau_\perp(n)$ be the squarefree extremal function of Erdős problem #1100. Erdős and Simonovits proved $(\sqrt{2} + o(1))^k < g_{\mathrm{sf}}(k) < (2-c)^k$. We give an exact combinatorial identity expressing $\tau_\perp$ as a count of *unblocked balanced splits*, and use it to reduce the problem, in a tight-band weight limit, to a sum of within-cardinality-layer adjacency counts $N(k,j)$. A first-moment analysis indicates $\mathbb{E}[N(k,j)] \asymp \binom{k-j}{j}$, so that $\sum_j \mathbb{E}[N(k,j)] \asymp \sum_j \binom{k-j}{j} = F_{k+1} \asymp \varphi^k$, the golden ratio. This *targets*

$$g_{\mathrm{sf}}(k) \;\geq\; (\varphi + o(1))^k, \qquad \varphi = \tfrac{1+\sqrt{5}}{2} = 1.61803\ldots,$$

raising the Erdős–Simonovits lower bound from base $\sqrt{2} = 1.41421\ldots$ to φ. The identity and the reduction are proved here, and the within-layer adjacency lemma is established in a companion note [4]. A further companion note [5] discharges the transfer from the tight-band model to prime weights — via a prime-block construction and two sequential limits — reducing it to that same within-layer lemma for *uniform* weights, with no base-measure generalization required. The golden-ratio bound is thus conditional only on that lemma, whose proof we believe to be complete but have not had independently refereed. Numerical evidence is consistent with the prediction. Combined with the unconditional upper bound $g_{\mathrm{sf}}(k) \leq (1.88988\ldots)^k$, proved in the companion note [6], this confines the growth rate $\gamma = \lim_k g_{\mathrm{sf}}(k)^{1/k}$ to $[\varphi,\, 1.88988\ldots]$, the lower endpoint inheriting the conditional status of the within-layer lemma above. The interval is non-degenerate, so γ remains undetermined; in particular it is open whether $\gamma = \varphi$ or $\gamma > \varphi$, the finite-k maxima exceeding the φ^k lower bound by bounded factors that do not by themselves force a larger base.

1 Background and results

Let $n = p_1 p_2 \cdots p_k$ be squarefree with $\omega(n) = k$, and let $1 = d_1 < d_2 < \cdots < d_{\tau(n)} = n$ be its divisors in increasing order. Define

$$\tau_\perp(n) = \#\{\, 1 \leq i < \tau(n) :\ \gcd(d_i, d_{i+1}) = 1 \,\},$$

the number of coprime adjacent divisor pairs, and the squarefree extremal function

$$g_{\mathrm{sf}}(k) = \max_{\substack{\omega(n)=k \\ n \text{ squarefree}}} \tau_\perp(n).$$

This is the function in part three of Erdős problem #1100 [1]; the function $\tau_\perp$ itself was introduced by Erdős and Hall [3]. The known bounds for $g_{\mathrm{sf}}(k)$, due to Erdős and Simonovits [2], are

$$(\sqrt{2} + o(1))^k \;<\; g_{\mathrm{sf}}(k) \;<\; (2-c)^k \tag{1}$$

for some constant $c > 0$. As noted in the problem's discussion, the proofs of (1) are not published in [2]; recreating and improving them is the motivation here. An explicit elementary upper bound $g_{\mathrm{sf}}(k) \leq (1.88988\ldots)^k$, i.e. $c = 0.110\ldots$, is given in the companion note [6].

The subset model. Since n is squarefree, divisors correspond to subsets $S \subseteq [k] = \{1, \dots, k\}$ via $d_S = \prod_{i \in S} p_i$, and $d_S < d_T \iff \sigma(S) < \sigma(T)$ where $\sigma(S) = \sum_{i \in S} a_i$, $a_i = \log p_i$. Coprimality is disjointness: $\gcd(d_S, d_T) = 1 \iff S \cap T = \varnothing$. More generally, for any $a_1, \dots, a_k > 0$ with distinct subset sums, order the 2^k subsets increasingly by σ and let $\tau_\perp$ count the adjacent pairs (S, T) with $S \cap T = \varnothing$. The prime case is $a_i = \log p_i$; passing between the two is discussed in Section 5.

Results. Section 2 proves an exact identity (Theorem 1) realizing $\tau_\perp$ as a count of unblocked balanced splits; this is the equality form of the "at most one per leftover set" bookkeeping behind the upper bound of [6]. Section 3 proves that, in the tight-band limit, $\tau_\perp$ decomposes over cardinality layers (Proposition 1). Section 4 carries out the first-moment heuristic leading to the golden ratio and states the conditional lower bound (Theorem 2). Section 5 states the remaining open step and the numerical evidence, and Section 6 records why φ is only a floor and how the method relates to the Erdős distinct-subset-sums literature.

2 An exact identity

Fix $a_1, \dots, a_k > 0$ with distinct subset sums and write $\sigma(S) = \sum_{i \in S} a_i$. For $K \subseteq [k]$, a *split* of K is an unordered partition $K = A \sqcup B$; its *imbalance* is $|\sigma(A) - \sigma(B)|$. Since subset sums are distinct, the *balanced split* $K = A_K \sqcup B_K$ minimizing imbalance is unique; write $\sigma(A_K) \le \sigma(B_K)$. Call K *unblocked* if no subset $R \subseteq [k]$ has $\sigma(R) \in \big(\sigma(A_K), \sigma(B_K)\big)$.

Theorem 1. *With the ordering above,*

$$\tau_\perp \;=\; \#\{\, K \subseteq [k] :\; K \text{ is unblocked} \,\}.$$

Moreover $K \mapsto \{A_K, B_K\}$ is a bijection from unblocked sets to coprime adjacent pairs, with inverse $\{S, T\} \mapsto S \cup T$.

Proof. Let (S, T) be a coprime adjacent pair, $\sigma(S) < \sigma(T)$, and put $K = S \cup T$ (a disjoint union, as $S \cap T = \varnothing$). Then S, T are complementary in K, so $\sigma(S) + \sigma(T) = \sigma(K)$ and hence $\sigma(S) < \frac{1}{2}\sigma(K) < \sigma(T)$. Adjacency means no subset of $[k]$ – in particular no subset of K – has σ-value in $\big(\sigma(S), \sigma(T)\big)$. Thus among subsets of K, S is the closest to $\frac{1}{2}\sigma(K)$ from below and T the closest from above. The balanced split minimizes $|\sigma(A) - \sigma(B)| = 2\big|\frac{1}{2}\sigma(K) - \sigma(A)\big|$, i.e. it picks the subset of K closest to $\frac{1}{2}\sigma(K)$; that subset is S (or T), so $\{A_K, B_K\} = \{S, T\}$ and K is unblocked.

Conversely, if K is unblocked then A_K, B_K are complementary in K, hence disjoint, and by definition no subset has σ-value strictly between $\sigma(A_K)$ and $\sigma(B_K)$, so $\{A_K, B_K\}$ is a coprime adjacent pair. The two maps are mutually inverse: $S \cup T = K$ recovers the ground set, and $\{A_K, B_K\} = \{S, T\}$ by the previous paragraph. The count follows. □

3 The layer reduction

Take $a_i = 1 + \varepsilon x_i$ with $x_1, \dots, x_k$ generic and $\varepsilon > 0$. Then $\sigma(S) = |S| + \varepsilon \sum_{i \in S} x_i$, and for ε small enough (depending on the x_i) the increasing σ-order is the lexicographic order by $\big(|S|,\ \sum_{i \in S} x_i\big)$: subsets are sorted first by cardinality, ties broken by the x-weight. Write $\tau_\perp^{\mathrm{lex}}$ for the resulting count and, for $0 \le j \le \lfloor k/2 \rfloor$,

$$N(k, j) = \#\big\{\text{coprime adjacent pairs } (S, T) :\; |S| = |T| = j\big\},$$

the disjoint adjacencies *within* cardinality layer j (disjointness forces $2j \le k$).

Proposition 1. *In the tight-band limit,*

$$\tau_{\perp}^{\mathrm{lex}} = \sum_{j=0}^{\lfloor k/2 \rfloor} N(k,j) \;+\; O(k),$$

and consequently $g_{\mathrm{sf}}(k) \geq \tau_{\perp}^{\mathrm{lex}} = \sum_j N(k,j) + O(k)$ *for a suitable squarefree* n.

Proof. In the lexicographic order, the size-j subsets occupy one contiguous block for each j. An adjacent pair therefore either lies within a single block (both of size j: counted by $N(k,j)$, and coprime only if $2j \leq k$) or straddles two consecutive blocks (sizes j and $j+1$). There is exactly one straddling pair per block boundary, hence k of them in total, contributing $O(k)$. Summing over blocks gives the displayed identity. The final inequality holds because $\tau_{\perp}^{\mathrm{lex}}$ is realized by a generic real weight vector, hence (after rescaling and approximation) by some squarefree n with $\omega(n) = k$; see Section 5. □

4 First moment and the golden ratio

We now model the x_i as i.i.d. with a bounded density (e.g. Uniform$[0,1]$), so that each layer becomes a random point configuration, and estimate $\mathbb{E}[N(k,j)]$.

Heuristic. The $\binom{k}{j}$ size-j sums have empirical spread σ_j with $\sigma_j^2 \asymp j(k-j)/k$ and local density $\rho_j \asymp \binom{k}{j}/\sigma_j$ near the center. A fixed disjoint pair (S,T) has gap $\Delta = \sigma(T) - \sigma(S)$ of scale $\sigma_\Delta \asymp \sqrt{j}$, so it is adjacent with probability of order $f_\Delta(0)/\rho_j \asymp \sigma_j/(\sigma_\Delta \binom{k}{j}) \asymp 1/\binom{k}{j}$. There are $\frac{1}{2}\binom{k}{j}\binom{k-j}{j}$ disjoint pairs, whence the $\binom{k}{j}$ cancels and

$$\mathbb{E}[N(k,j)] \;\asymp\; \binom{k-j}{j}. \tag{2}$$

The peak layer is $j \approx 0.276\,k$ (the saddle of $\binom{k-j}{j}$); the middle layer $j = k/2$ contributes only $O(1)$. Summing (2) over layers and using the Fibonacci identity $\sum_{j\geq 0}\binom{k-j}{j} = F_{k+1}$,

$$\mathbb{E}\big[\tau_{\perp}^{\mathrm{lex}}\big] = \sum_j \mathbb{E}[N(k,j)] + O(k) \;\asymp\; F_{k+1} \;\asymp\; \frac{\varphi^{k+1}}{\sqrt{5}}. \tag{3}$$

A rigorous lower bound $\mathbb{E}[N(k,j)] \gg \binom{k-j}{j}$ underlying (2) is proved in the companion within-layer lemma [4]; the proof is elementary (two one-dimensional density bounds for sums of uniforms, a codimension-two small-ball bound that renders the "swap-type" internal blockers exponentially negligible, and a hypergeometric concentration estimate via a Hoeffding tail bound). Granting the within-layer lemma [4] and the transfer of Section 5, $g_{\mathrm{sf}}(k) \geq \mathbb{E}[\tau_{\perp}^{\mathrm{lex}}]$ yields the following.

Theorem 2 (conditional)**.** *The transfer from tight-band to prime weights is carried out in the companion note [5], where a prime-block construction with two sequential limits reduces it to the within-layer estimate* $\mathbb{E}[N(k,j)] \gg \binom{k-j}{j}$ *(uniform for* $j \asymp 0.276\,k$*) for* uniform *weights — exactly the estimate proved in [4]. Granting that lemma, then*

$$g_{\mathrm{sf}}(k) \;\geq\; (\varphi + o(1))^k, \qquad \varphi = \tfrac{1+\sqrt{5}}{2},$$

raising the Erdős–Simonovits lower bound in (1) *from base* $\sqrt{2}$ *to* φ.

5 What is proved and what awaits independent verification

Proof architecture. The program spans five notes, including this one. Figure 1 shows their dependence, and the status divides into three tiers plus the open question.

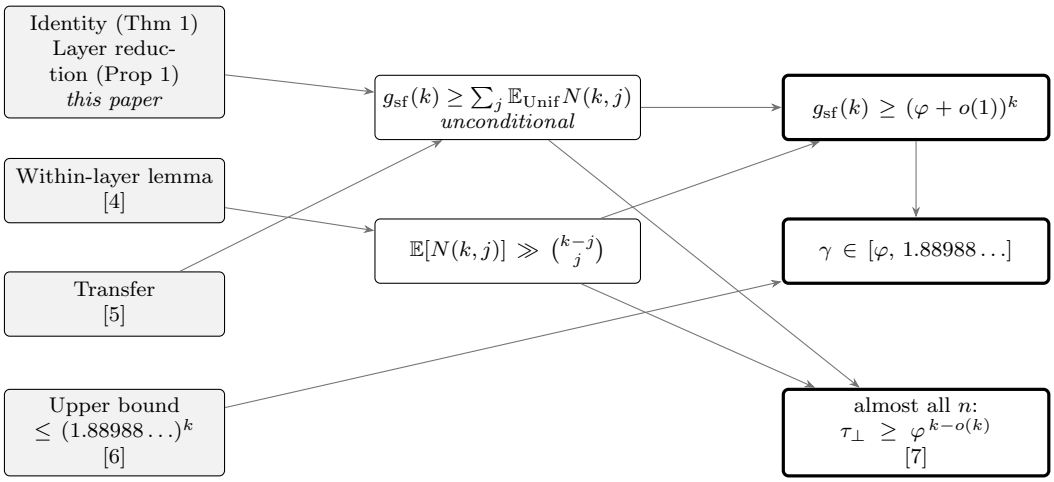


Figure 1: Dependence of the program. Shaded boxes are sources; the middle column is the unconditional reduction; the three bold boxes are the endpoints — the conditional golden-ratio floor (Theorem 2), the growth-rate interval, and the typicality statement of [7].

Unconditional. Theorem 1 (the identity) and Proposition 1 (the layer reduction) give $\tau_\perp^{\mathrm{lex}} = \sum_j N(k,j) + O(k)$; the transfer note [5] discharges the passage to prime weights, yielding $g_{\mathrm{sf}}(k) \ge \sum_{j \le k/2} \mathbb{E}_{\mathrm{Unif}}[N(k,j)]$ on fixed-proportion PNT alone; and [6] gives $g_{\mathrm{sf}}(k) \le (1.88988\ldots)^k$.

Conditional on the within-layer lemma [4] (proof complete, not independently refereed). The lemma supplies $\mathbb{E}[N(k,j)] \gg \binom{k-j}{j}$; with the transfer and $\sum_j \binom{k-j}{j} = F_{k+1}$ this gives $g_{\mathrm{sf}}(k) \ge (\varphi + o(1))^k$ (Theorem 2), raising the proven floor from $\sqrt{2}$ to φ.

Conditional on [4] plus a standard multivariate local limit. The companion note [7] shows the candidate and blocked within-layer counts concentrate, so almost all n in the prime-block ensemble of [5] satisfy $\tau_\perp(n) \ge \varphi^{k-o(k)}$: the golden-ratio floor is the *typical* value, not merely the maximum.

Open. The growth rate $\gamma = \lim_k g_{\mathrm{sf}}(k)^{1/k}$ lies in $[\varphi, 1.88988\ldots]$; whether $\gamma = \varphi$ is undetermined and routes through lowering the upper bound to meet the floor.

The remainder of this section elaborates the two conditional tiers.

Theorem 1 and Proposition 1 are proved unconditionally, and the within-layer estimate $\mathbb{E}[N(k,j)] \gg \binom{k-j}{j}$ (uniform for $j \asymp 0.276\,k$) is proved in the companion note [4]. That proof reduces, via a truncated first moment, to two elementary one-dimensional density bounds for sums of uniforms – (i) $\mathbb{P}[|\Delta| \le \theta] = 2\theta f_\Delta(0)(1 + o(1))$ with $f_\Delta(0) = \|f_{\mathrm{IH}_j}\|_2^2 \ge c_0/\sqrt{j}$ (from continuity of the Irwin–Hall self-convolution and Chebyshev), and (ii) a conditional density bound $\sup_w f_{X(R)|X(S),X(T)}(w) \le A/(s\sqrt{m})$, $m = |R \setminus (S \cup T)|$ (by convolution smoothing) – together with (iii) the hypergeometric concentration

$$\sum_{m \ge 1} \binom{2j}{j-m} \binom{k-2j}{m} m^{-1/2} \;\le\; C \binom{k}{j} / \sqrt{j},$$

formerly verified only numerically and now established by an elementary Hoeffding tail bound for the hypergeometric law of m (with sharp constant $C \to 1/\sqrt{1-2\beta}$); and (iv) a codimension-two small-ball bound for the overlap-free ($m = 0$) internal blockers — the swap-type neighbors — which, writing the gap as a sum $\Delta = U + V$ of two independent signed-sum forms, forces both into a window of width θ and so costs θ^2 rather than θ, rendering that whole class an exponentially small fraction of the main term. (Ingredient (iv) replaces an incorrect "discarded by counting" argument in the first version of [4], identified in external review.) The lower bound of Theorem 2 therefore rests on the transfer, carried out in the companion note [5] and summarized here:

> **The transfer.** The estimate is for i.i.d. tight-band weights; one needs the i.i.d. mean of $\tau_\perp^{\mathrm{lex}}$ to be attained (up to the exponential rate) by prime weights $a_i = \log p_i$. Since $\tau_\perp$ depends only on the order type of the subset sums and $g_{\mathrm{sf}}(k)$ is a maximum over prime vectors, this is a question of realizing the relevant order types by logarithms of primes. The companion note [5] settles it by a prime-block construction with two sequential limits: the k primes are drawn from a block $[Q, (1+\eta)Q]$, and one takes $Q \to \infty$ at fixed proportion η — whereupon the divisor order refines exactly by cardinality and, by the prime number theorem, the weights $\log(p/Q)$ acquire an explicit limit law — and then $\eta \to 0$, which sends that law, suitably rescaled, to the *uniform* law in total variation. The finite-pool, equidistribution, and base-measure corrections all vanish in the limits, and because $\eta \to 0$ is taken after $Q \to \infty$ only ordinary (fixed-proportion) PNT is used, with no short-interval input. The transfer thereby reduces to the within-layer estimate for uniform weights, so the single residual is the uniform lemma [4] itself, whose proof we believe complete but which has not yet been independently refereed.

Numerical evidence. The identity of Theorem 1 checks exactly over all K in exact arithmetic for $k \le 10$. The mean of $\tau_\perp$ over i.i.d. weights is exponential, with base rising from $\approx$1.55 at $k = 8$ to $\approx$1.56 at $k = 22$ – well below φ and still climbing – consistent with the slow ($\sim k^{-1/4}$ prefactor) convergence to φ predicted by (3); the per-layer counts track $\binom{k-j}{j}$ up to a bounded factor across the bulk layers; and the mean and maximum nearly coincide, so the phenomenon is typical rather than a rare event. This last point — that the floor is attained typically and not only by the maximizer — is made rigorous in the companion note [7], where the within-layer count is shown to concentrate.

6 Remarks

Remark 1 (φ is a floor)**.** *Equation* (3) *is the* mean *over the tight-band ensemble, hence — with the transfer — a lower bound on $g_{\mathrm{sf}}(k)$ of base φ. It need not be the growth rate: the finite-k maxima exceed it, e.g. the best n found at $k = 10$ has $\tau_\perp \ge 133$, against the lower bound $F_{11} = 89$ and against $\varphi^{10} \approx 123$. This shows the bound is not tight at finite k, but the excess is a* bounded prefactor*, not a larger exponential base: across the searched range $k \le 10$ the ratio $g_{\mathrm{sf}}(k)/\varphi^k$ stays in $[1.08, 1.24]$ and does not grow, so the data are equally consistent with $g_{\mathrm{sf}}(k) \asymp \varphi^k$ and with a strictly larger rate. (The comparison "293 at $k = 12$ against $F_{13} = 233$" overstates the case: $F_{13} = \varphi^{13}/\sqrt{5}$ carries a $1/\sqrt{5}$ factor, and against $\varphi^{12} \approx 322$ the value 293 shows no excess.) With the unconditional upper bound $(1.88988\ldots)^k$ of [6], the growth rate $\gamma = \lim_k g_{\mathrm{sf}}(k)^{1/k}$ satisfies $\varphi \le \gamma \le 1.88988\ldots$, the lower bound conditional on the within-layer lemma; its value — including whether $\gamma = \varphi$ — is open. The present contribution is to raise the proven floor from $\sqrt{2}$ to φ.*

Remark 2 (Relation to distinct subset sums)**.** *The analytic core – small-ball control of signed subset sums and binomial-coefficient bounds via anti-concentration – is the toolkit of the Erdős distinct-subset-sums problem; see Dubroff, Fox and Xu [8] and Steinerberger [9]. Those works*

bound the largest element of a sum-distinct set and are formally unrelated to $g_{\mathrm{sf}}(k)$, but the methods transfer. We note that a vertex-isoperimetric (Harper/Littlewood–Offord) argument, central there, does not *shortcut the within-layer estimate: it produces close pairs differing in a single coordinate, whereas the pairs counted by $N(k, j)$ are disjoint, i.e. maximally far in Hamming distance.*

Computation. Supporting code is consolidated in a single Python script at `https://github.com/michaelmross/Erdos/tree/main/1100`.

Acknowledgments

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